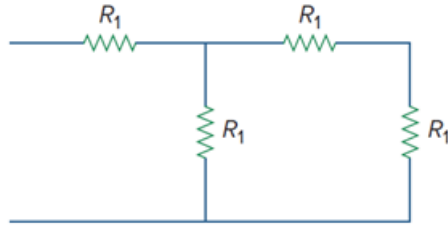


Problem

Given the circuit in Fig. 2.101 and that the resistance, R_{eq} , looking into the circuit from the left is equal to $100\ \Omega$, determine the value of R_1 .

Figure 2.101:



Step-by-step solution

Step 1 of 2

Refer to circuit diagram in Figure 2.101 in the textbook.

The equivalent resistance of the circuit seen from left is,

$$\begin{aligned} R_{eq} &= [(R_1 + R_1) \parallel R_1] + R_1 \\ &= [2R_1 \parallel R_1] + R_1 \\ &= \frac{(2R_1)(R_1)}{3R_1} + R_1 \\ &= \frac{5R_1}{3} \end{aligned}$$

Step 2 of 2

The equivalent resistance value is $100\ \Omega$.

Calculate the resistor value.

$$\begin{aligned} \frac{5R_1}{3} &= 100 \\ R_1 &= \frac{3(100\ \Omega)}{5} \\ &= 60\ \Omega \end{aligned}$$

Therefore, the value of resistance, R_1 is $60\ \Omega$.